MIL-M-28787/27A(EC)
14 May 1979
SUPERSEDING
MIL-M-28787/27(NAVY)
24 October 1973

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COUNTER, UP AND DOWN, BINARY, DIGITAL, KEY CODE GDM

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic binary UP and DOWN, counter module.

1.2 Module description. The following electrical parameters reflect 0° to 60°C the end-of-life limits from table II.

Name	Counter, up and down, binary
Family	Digital
Key code	GDM
Size	1A
Weight	35 grams maximum
Failure rate	1.88 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Supply voltage	5 Vdc ±10%
Maximum power dissipation	1.85 W (outputs not loaded)
Clock inputs	
Counting frequency	DC to 20 MHz maximum
Pulse width (high level)	20 ns minimum
Pulse negative slope	100 ns maximum
Triggering edge	Negative
Input threshold voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 5043, Department of the Navy, Washington, D.C. 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Input current	
High level (V_{IH} = 4.5 V)	
Master set	0 µA minimum, 800 µA maximum
Master reset	0 µA minimum, 800 µA maximum
Carry-in	0 µA minimum, 120 µA maximum
Count enable	0 µA minimum, 40 µA maximum
Clock	0 µA minimum, 40 µA maximum
Up and down	0 µA minimum, 40 µA maximum
Input current	
Low level (V_{IL} = 0 V)	
Master set	1.0 mA minimum, 29.00 mA maximum
Master reset	1.0 mA minimum, 29.00 mA maximum
Carry-in	1.0 mA minimum, 5.43 mA maximum
Count enable	0.1 mA minimum, 1.81 mA maximum
Clock	0.1 mA minimum, 1.81 mA maximum
Up and down	0.1 mA minimum, 1.81 mA maximum
Output voltage	
High level	2.4 V minimum
Low level	0.4 V maximum
Output drive	
High level (V_{OH} = 2.4 V)	800 µA minimum
Low level (V_{OL} = 0 V)	10.4 mA maximum
Output short circuit current	20 mA minimum, 74 mA maximum
Propagation delay time (clock to Q):	
t_{PHL}	5 ns minimum, 50 ns maximum
t_{PLH}	5 ns minimum, 45 ns maximum
Propagation delay time (carry-in to carry-out):	
t_{PHL}	5 ns minimum, 35 ns maximum
t_{PLH}	5 ns minimum, 35 ns maximum
Master set delay time	5 ns minimum, 40 ns maximum
Master reset delay time	5 ns minimum, 45 ns maximum
Master set and reset pulse width (low level)	25 ns minimum
Setup time (before negative transition of clock):	
Carry-in	25 ns minimum
Count enable	25 ns minimum
Up /down	25 ns minimum
Hold time (after negative transition of clock):	
Carry-in	2 ns minimum
Count enable	2 ns minimum
Up/down	2 ns minimum

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1.3 Module functional requirements. This module contains 4 four-bit synchronous binary UP and DOWN counters. Each four-bit counter has a carry-in or data input (CI), count enable input (CE) clock pulse input (CP), UP/$\overline{\text{DOWN}}$ count input (U/D), carry-out or terminal count output ($\overline{\text{CO}}$) and four true outputs (Qa, Qb, Qc, and Qd). All four counters are controlled by one master set ($\overline{\text{MS}}$) and one master reset (MR). The entry and propagation of data is performed in a synchronous manner with the clock, which is active on its negative excursion. The input from a previous stage or other source is channeled through carry-in and its propagation can be inhibited by the count enable. The direction of the counter is steered from the UP/$\overline{\text{DOWN}}$ input, where a low level will cause a down count and a high level will accomplish an up count. The output code of the up and down counter is a weighted binary code (8421) and the output sequence generated is the binary equivalent of the decimal numbers 0 through 15. The set and reset on the binary elements provide a synchronous entry with respect to the clock line. A reset low level will cause a count of "0" (0000) and a set low level will accomplish a count of 15 (1111). If all four output bits and the up and down and carry-in inputs are at a high level or if all four output bits and the up and down inputs are at a low level and the carry-in input is at a high level, the carry-out (terminal count) output will be at a high level. The carry-out level corresponds to the following Boolean expression.

$$\text{Carry-out} = \text{Carry-in}\ (Qa, Qb, Qc, Qd, Up + \overline{Qa}, \overline{Qb}, \overline{Qc}, \overline{Qd}, \overline{DOWN})$$

1.3.1 Input and output relationship. Each four bit UP and DOWN binary counter has the input to output relationship of figure 1 and table I for positive logic.

1.3.2 Parallel outputs. Paralleling of outputs is not permissible.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.88 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table II and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

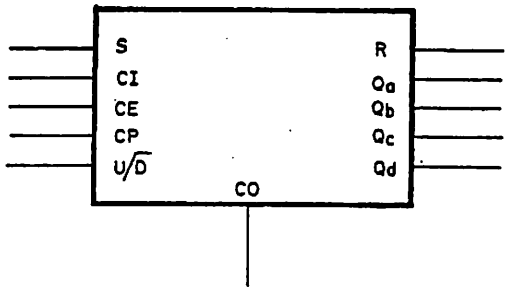
3.5.1 Key code. The module key code shall be GDM.

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TABLE I. Mode of operation.

	Inputs					Output function				
	Set	Reset	Carry -in	Count enable	Up/down	Qa	Qb	Qc	Qd	Decimal
Asynchron-ous Operation	H	L	X	X	X	L	L	L	L	0
	L	H	X	X	X	H	H	H	H	15
Synchron-ous Operation (Note "1")	H	H	L	X	X	Hold				
	H	H	X	L	X	Hold				
	H	H	H	H	L	"Down" count				
	H	H	H	H	H	"Up" count				

NOTES:
1. Function is synchronous with negative going transition of the clock.
2. X = Don't care term.
3. Carry-out = Carry-in (Qa, Qb, Qc, Qd, UP + $\overline{Qa}$, $\overline{Qb}$, $\overline{Qc}$, $\overline{Qd}$, $\overline{DOWN}$.



PIN DESIGNATIONS:

CI = carry-in/data input
CE = count enable input
CP = clock pulse input
U/$\overline{D}$ = Up/$\overline{Down}$ count input
S = set input
R = reset input
CO = carry-out/terminal count output
Q = binary output (Qd, Qc, Qb, Qa = 8, 4, 2, 1)

FIGURE 1. Data input.

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3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/27-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 2.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table II.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table II (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table II for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below:

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	3
Pulse generator	Item J01	2
DC voltmeter	Item V03	1
DC ammeter	Item X01	1
Oscilloscope	Item D05	1
Oscilloscope probe	Item Y01	2
Megohm bridge	Item R06	1
Capacitance meter	Item C05	1
Resistors		
50Ω, ±5%, 1 W		3
523Ω ±1%, 1/8 W		20
Value as required for figure 9, note 4, 1/8 W		20
412Ω ±1%, 1/8 W		1
3.32 kΩ ±1%, 1/8 W		20
Capacitors		
Value as required for figure 9, note 2		20
Value as required for figure 9, note 6		20

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Switches	No. required (maximum per test figure)
SPST	2
DPST	1
Diode	
1N4150, or equivalent	20
Component	
Logic state indicators as required for figure 9, note 5 (DIALCO 903-1458-1431-523, or equivalent)	5

All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table II.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through GG shall be performed in any order with the following exceptions. If test condition A is part of a sequence of tests, the order of testing shall be as follows: A where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through GG in table II shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through CC, FF, and GG shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Transient thermal test. The transient thermal test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 4.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 5. Following the vibration tests, the electrical tests of table II which are marked with an (X), and test condition AA, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests B through CC, FF, and GG shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table II. Following the return of the module to 25°C, electrical tests B through GG shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table II.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

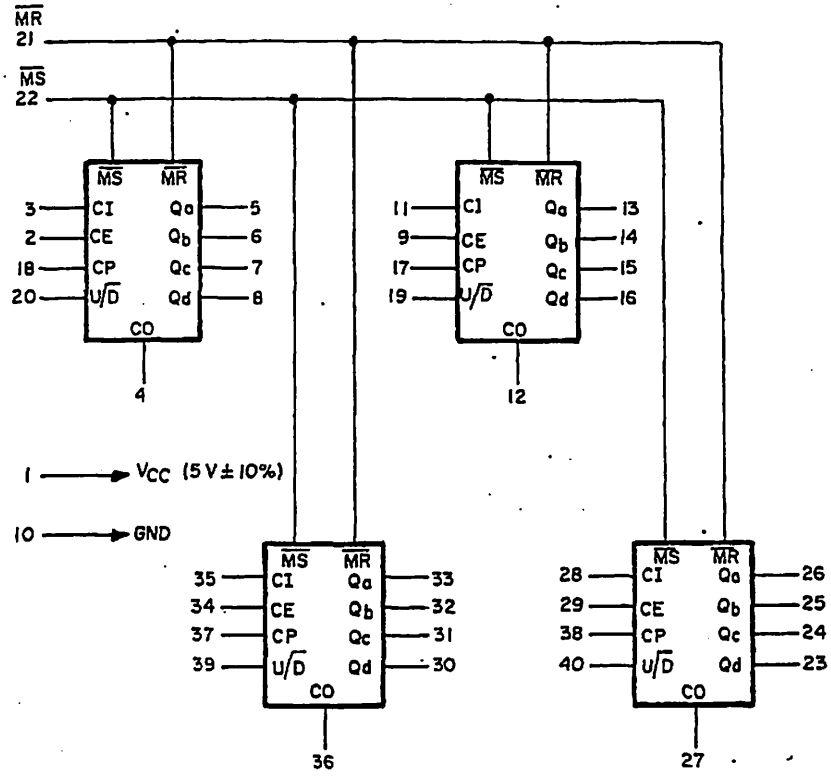
6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table II, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference or classification. Part number M28787/27-1 supersedes part number M28787/27. The two parts are interchangeable.

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PIN DESIGNATIONS:

CI = carry-in/data input.
CE = count enable input
CP = clock pulse input
CO = carry-out or terminal count output

COUNT INPUT

$\overline{U/D}$ = Up/$\overline{down}$
$\overline{MS}$ = master set input
$\overline{MR}$ = master reset input
Q = Binary output (Qd, Qc, Qb, Qa = 8, 4, 2, 1).

FIGURE 2. GDM module function diagram, quad four bit synchronous binary up/down counter.

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TABLE II. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ± 5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-20	Output voltage (low loaded)	III	S	0.1	0.8	0.1	0.8	0.1	0.8	0.1	0.8	V
B1-2	Master reset and set input current, low level	IV	X	1.0	29.0	1.0	29.0	1.0	29.0	1.0	29.0	mA
B3-6	Carry-in input current, low level	IV	X	1.0	5.43	1.0	5.43	1.0	5.43	1.0	5.43	mA
B7-18	Count enable, clock and U/$\overline{D}$ input current, low level	IV	X	0.1	1.81	0.1	1.81	0.1	1.81	0.1	1.81	mA
C1-2	Master reset and set input current, high level	IV	X	0	800	0	800	0	800	0	800	µA
C3-6	Carry-in input current, high level	IV	X	0	120	0	120	0	120	0	120	µA
C7-18	Count enable, clock and U/D input current, high level	IV	X	0	40	0	40	0	40	0	40	µA
D1-20	Output short circuit current	IV	X	20	74	20	74	20	74	20	74	mA
E1-20	Output voltage, high level	V	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-20	Output voltage, low level	V	X	0.00	0.40	0.00	0.42	0.00	0.42	0.00	0.44	V
G1-4	Functional count up check	V	X	← See note 6, figure 8 →								
H1-4	Functional count down check	V	X	← See note 6, figure 8 →								
J1-2	Power supply current	VI	X	50	336	50	336	50	336	50	336	mA
K1-16	Transition time (t_{THL})	VII	S	2	20	2	20	2	25	2	25	ns

See notes at end of table.

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TABLE II. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ± 5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L1-16	Propagation delay time (t_{PHL})	VII	S	5	45	5	45	5	50	5	50	ns
M1-16	Transition time (t_{TLH})	VII	S	4	40	4	40	4	45	4	45	ns
N1-16	Propagation delay time (t_{PLH})	VII	S	5	40	5	40	5	40	5	45	ns
P1-16	Master set delay time	VIII	S	5	35	5	35	5	40	5	40	ns
Q1-16	Master reset delay time	VIII	S	5	40	5	40	5	45	5	45	ns
R1-4	Carry-out transition time, (t_{TLH})	IX	S	4	40	4	40	4	45	4	45	ns
S1-4	Carry-out transition time, (t_{THL})	IX	S	2	20	2	20	2	20	2	20	ns
T1-4	Carry-out delay time (t_{PHL})	IX	S	5	30	5	30	5	35	5	35	ns
U1-4	Carry-out delay time (t_{PLH})	IX	S	5	30	5	30	5	35	5	35	ns
V1-4	Carry-in setup and hold time check	X	S	765	845	765	845	765	845	765	845	ns
W1-4	Count enable setup and hold time check	X	S	765	845	765	845	765	845	765	845	ns
X1-4	Count up setup and hold time check (QA)	X	S	55	145	55	145	50	150	50	150	ns
X5-8	Count up setup and hold time check (QB)	X	S	155	245	155	245	150	250	150	250	ns
X9-12	Count up setup and hold time (QC)	X	S	355	445	355	445	350	450	350	450	ns

See notes at end of table.

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TABLE II. Module electrical requirements -Continued.

Test condition	Parameter	Ref test table	Test type (see notes)	Acceptance requirements 25° ± 5°C Ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
X13-16	Count up setup and hold time check (QD)	X	S	755	845	755	845	750	850	750	850	ns
Y1-4	Count down setup and hold time check (QA)	X	S	55	145	55	145	50	150	50	150	ns
Y5-8	Count down setup and hold time check (QB)	X	S	155	245	155	245	150	250	150	250	ns
Y9-12	Count down setup and hold time check (QC)	X	S	355	445	355	445	350	450	350	450	ns
Y13-16	Count down setup and hold time check (QD)	X	S	755	845	755	845	750	850	750	850	ns
Z	Isolation	XI	S	10	∞	10	∞	10	∞	10	∞	MΩ
AA	Power supply transient rise time	XII	S	2.5	7.5	2.5	7.5	2.5	7.5	2.5	7.5	µs
BB1-20	Output crosstalk, high level	XIII	Q	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
CC1-20	Output crosstalk, low level	XIII	Q	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
DD	Power supply transient amplitude	XIV	Q	0	400	0	400	0	400	0	400	mV
EE1-2	$\overline{MR}$ & $\overline{MS}$ input capacitance	XV	Q	16	96	16	96	16	96	16	96	pF
EE3-18	CI, CE, CP and U/$\overline{D}$ input capacitance	XV	Q	2	15	2	15	2	15	2	15	pF
FF1-4	Functional count up check	XVI	Q	← See note 5, figure 19 →								
GG1-4	Functional count down check	XVI	Q	← See note 5, figure 19 →								

See notes at end of table.

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TABLE II. Module electrical requirements -Continued.

NOTES:

1. All (X) parameters shall be 100 percent tested on all modules. Modules which have not previously been used in an operating system, or have been in storage for less than 2 years, shall conform to the 25° ±5°C initial acceptance requirements of table II.

2. All (S) parameters shall be 25°C ambient sample acceptance tested (1 percent or two (2) modules from each inspection lot whichever is greater).

3. All (Q) parameters shall be qualification tested only.

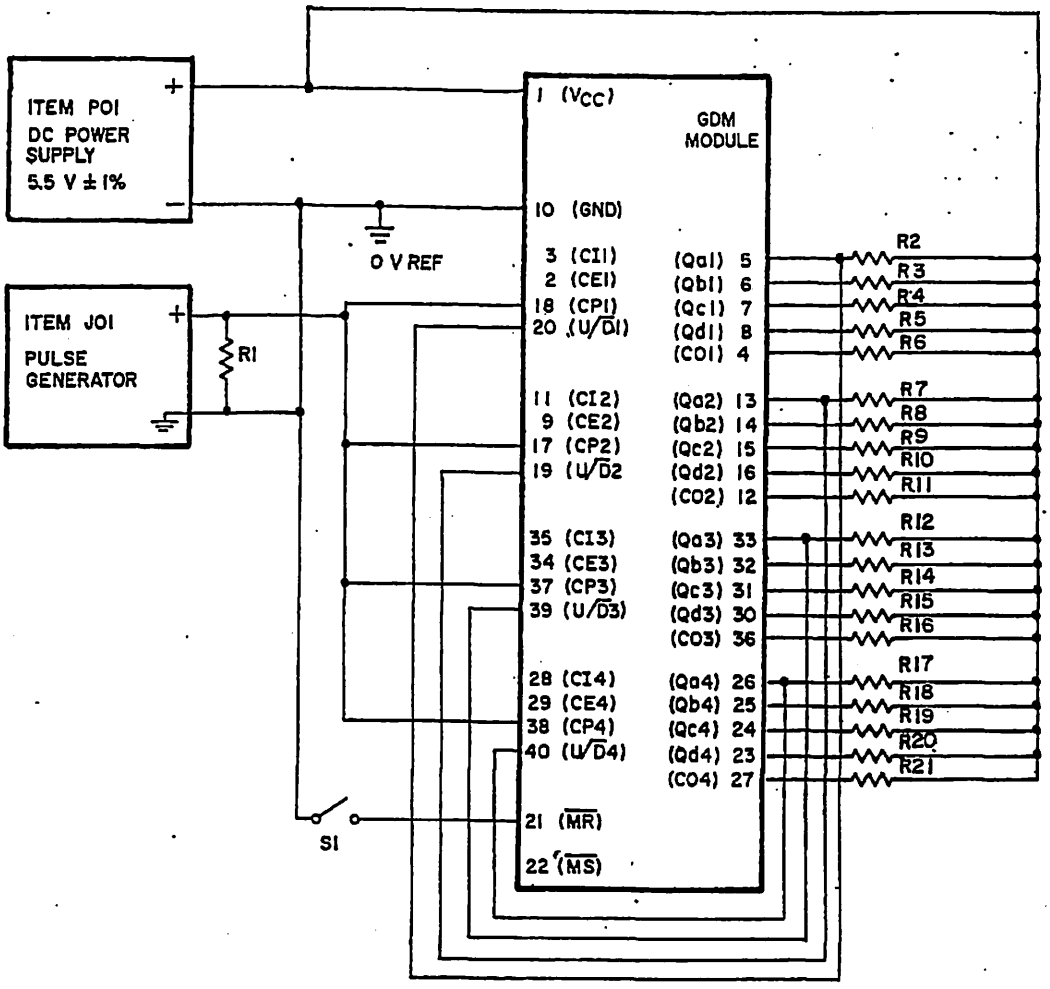
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Preparing activity:
Navy - EC

Agent:
DLA - ES

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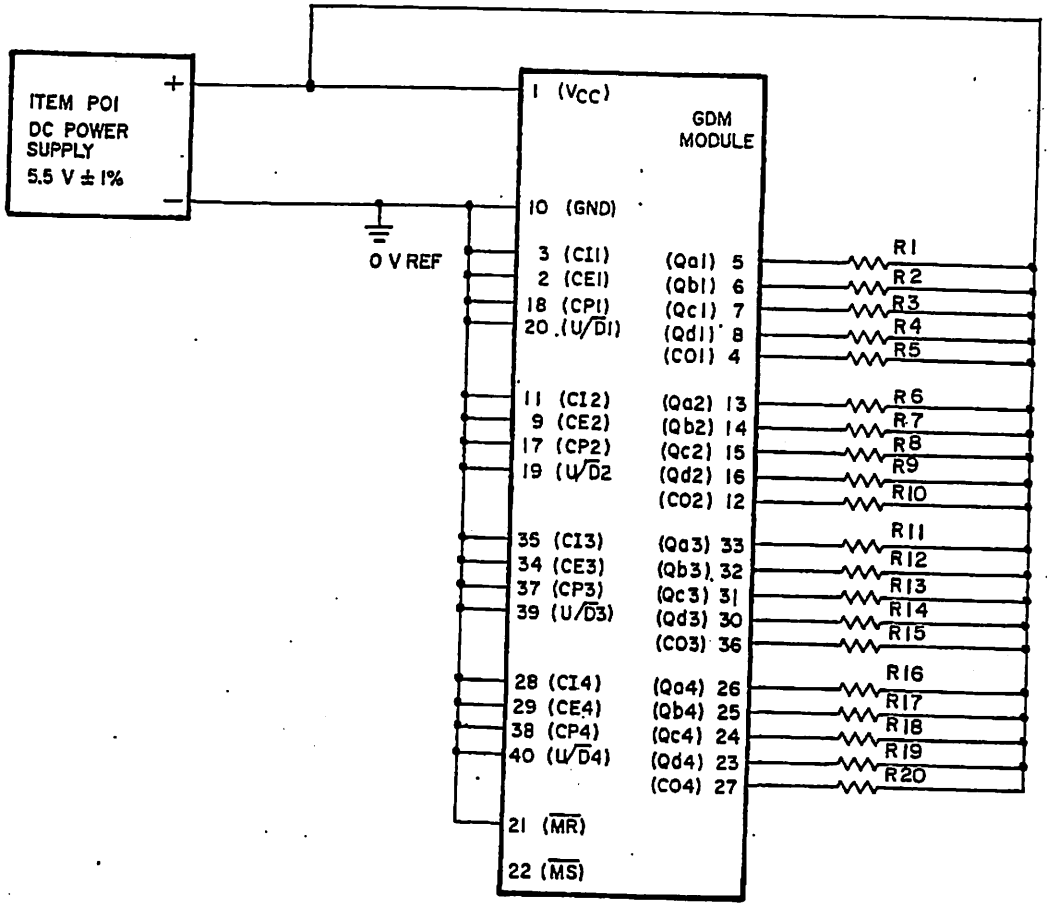
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. R2-R21 = 523Ω ±1%, 1/8 W.
3. Input pulse parameters:
   PW = 1 ±0.1 µs.
   PRP = 2 ±0.2 µs.
   $t_{TLH} = t_{THL}$ = 50 ±40 ns.
   PA = 4 ±0.5 V.
4. After making all circuit connections, close switch S1 for a minimum of 100 ns then open. Each output should then change state with each clock pulse.

FIGURE 3. Operating temperature and life test circuit.

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NOTE: R1-R20 = 523Ω ±1%, 1/8 W.

FIGURE 4. Transient thermal test circuit.

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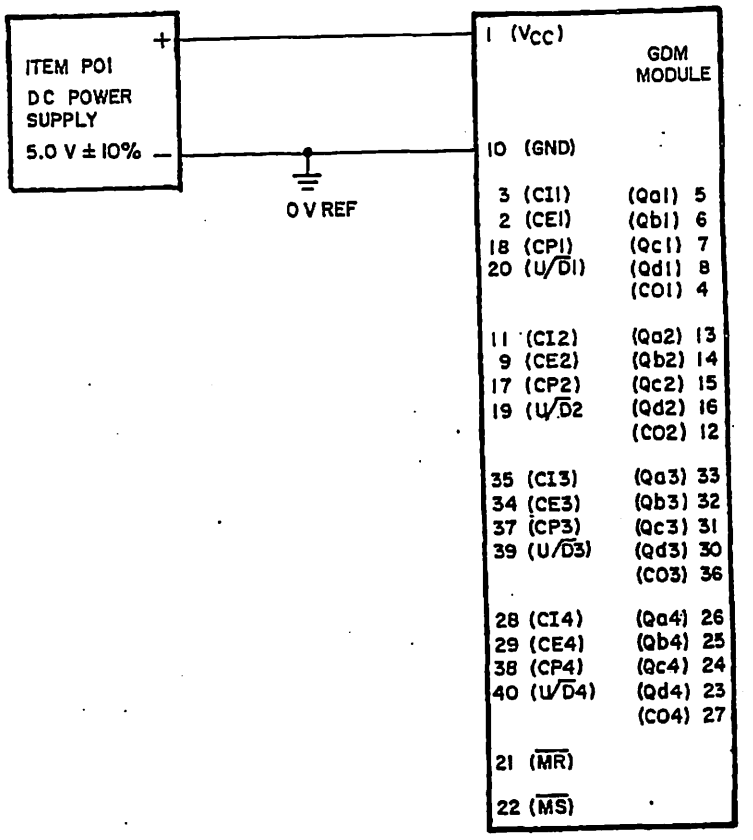


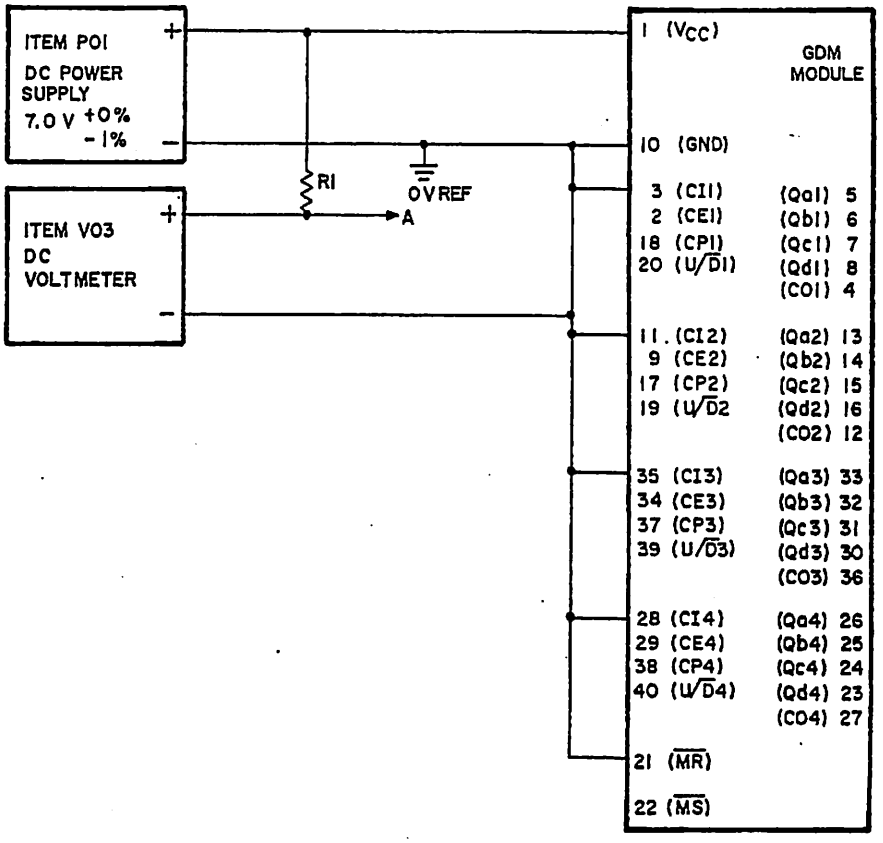
FIGURE 5. Vibration test circuit.

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
A1	QA1 V_{OUT} (low-loaded)	5	0.1	0.8	V	6
A2	QB1	6				
A3	QC1	7				
A4	QD1	8				
A5	CO1	4				
A6	QA2	13				
A7	QB2	14				
A8	QC2	15				
A9	QD2	16				
A10	CO2	12				
A11	QA3	33				
A12	QB3	32				
A13	QC3	31				
A14	QD3	30				
A15	CO3	36				
A16	QA4	26				
A17	QB4	25				
A18	QC4	24				
A19	QD4	23	↓	↓	↓	↓
A20	CO4 V_{OUT} (low-loaded)	27	0.1	0.8	V	6

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NOTE: R1 = 523Ω ±1%, 1/8 W.

FIGURE 6. V_{OUT} (low-loaded) test circuit for table III.

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TABLE IV. Input current (low-level and high-level) and output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Pin 10 to pin	Min	Max		
B1	$\overline{MR}$ Input current, low level	Open	21	Open	1.0	29.0	mA	7
B2	$\overline{MS}$		22	Open		29.0		
B3	CI1		3	20,22		5.43		
B4	CI2		11	19,22				
B5	CI3		35	39,22				
B6	CI4		28	40,22	1.0	5.43		
B7	CE1		2	Open	0.1	1.81		
B8	CP1		18					
B9	U/$\overline{D1}$		20					
B10	CE2		9					
B11	CP2		17					
B12	U/D2		19					
B13	CE3		34					
B14	CP3		37					
B15	U/$\overline{D3}$		39					
B16	CE4		29					
B17	CP4		38					
B18	U/$\overline{D4}$ Input current, low level	Open	40	Open	0.1	1.81	mA	7

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TABLE IV. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Pin 10 to pin	Min	Max		
C1	$\overline{MR}$ Input current, high level	21	Open	2,3,9,11, 28,29,34, 35,22	0	800	µA	7
C2	$\overline{MS}$	22		2,3,9,11, 28,29,34, 35,21		800		
C3	CI1	3		2,20,21		120		
C4	CI2	11		9,19,21				
C5	CI3	35		34,39,21				
C6	CI4	28		29,40,21		120		
C7	CE1	2		3,21		40		
C8	CP1	18		21				
C9	U/$\overline{D1}$	20		21				
C10	CE2	9		11,21				
C11	CP2	17		21				
C12	U/$\overline{D2}$	19		21				
C13	CE3	34		35,21				
C14	CP3	37		21				
C15	U/$\overline{D3}$	39		21				
C16	CE4	29		28,21				
C17	CP4	38		21				
C18	U/$\overline{D4}$ Input current, high level	40	Open	21	0	40	µA	7

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TABLE IV. Input current (low-level and high-level) and output short circuit current test - Continued.

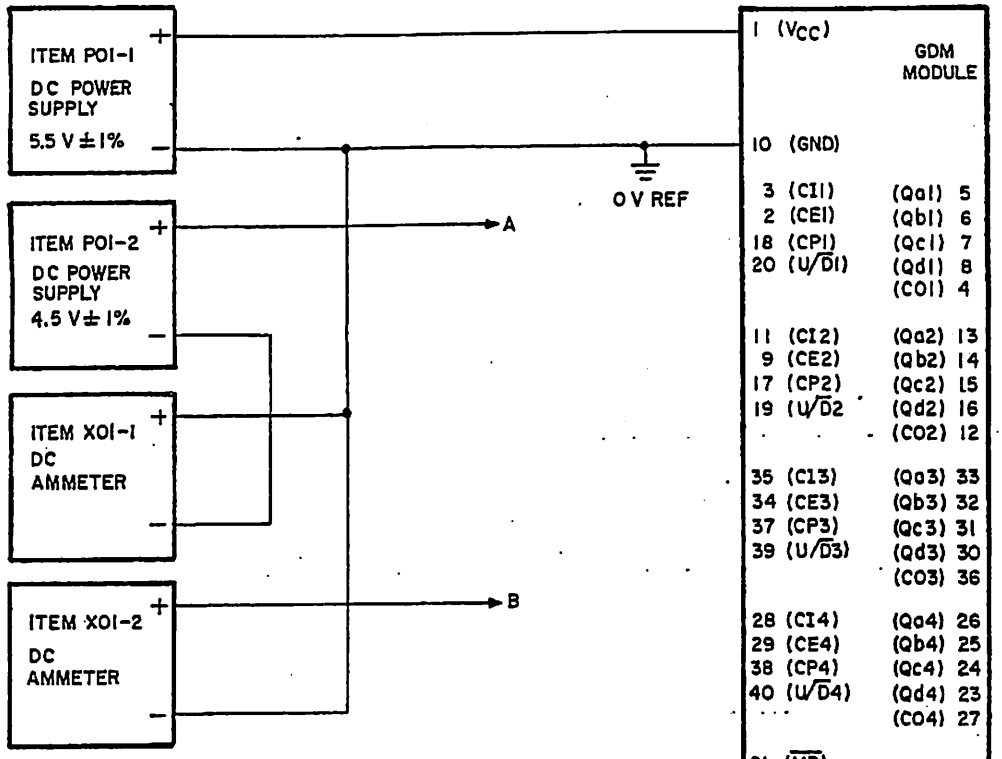
Test condition	Parameter	Connect.			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Pin 10 to pin	Min	Max		
D1	QA1 Short circuit current	Open	5	22	20	74	mA	7
D2	QB1		6					
D3	QC1		7					
D4	QD1		8					
D5	CO1		4					
D6	QA2		13					
D7	QB2		14					
D8	QC2		15					
D9	QD2		16					
D10	CO2		12					
D11	QA3		33					
D12	QB3		32					
D13	QC3		31					
D14	QD3		30					
D15	CO3		36					
D16	QA4 Short circuit current	Open	26	22	20	74	mA	7

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TABLE IV. Input current (low-level and high-level) and output short circuit current test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Pin 10 to pin	Min	Max		
D17	QB4 Short circuit current	Open	25	22	20	74	mA	7
D18	QC4	↓	24	↓	↓	↓	↓	↓
D19	QD4	↓	23	↓	↓	↓	↓	↓
D20	CO4 Short circuit current	Open	27	22	20	74	mA	7

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since ammeters X01-1 and X01-2 are not used simultaneously, only one ammeter X01 is required.
2. Only one output shall be shorted at a time, and the measurement taken within 10 seconds.

FIGURE 7. Input current (low-level and high-level) and output short circuit current test circuit for table IV.

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TABLE V. Output voltage (low-level and high-level), functional count up and count down check (input threshold conditions) test.

Test condition	Parameter	Connect						Initial limits 25° ±5°C Ambient		Unit	Test figure
		Pin 1 to pin	A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	QA1 Output voltage, high level MS(TH-L); MR(TH-H)	Open	Open	21	22	5	Open	2.4	4.5	V	8
E2	QB1			21		6					
E3	QC1			21		7					
E4	QD1			21		8					
E5	CO1			3,20, 21		4					
E6	QA2			21		13					
E7	QB2			21		14					
E8	QC2			21		15					
E9	QD2			21		16					
E10	CO2			11,19, 21		12					
E11	QA3			21		33					
E12	QB3			21		32					
E13	QC3			21		31					
E14	QD3			21		30					
E15	CO3			35,39, 21		36					
E16	QA4			21		26					
E17	QB4 Output voltage, high level MS(TH-L); MR(TH-H)	Open	Open	21	22	25	Open	2.4	4.5	V	8

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TABLE V. Output voltage (low-level and high-level), functional count up and count down check (input threshold conditions) test - Continued.

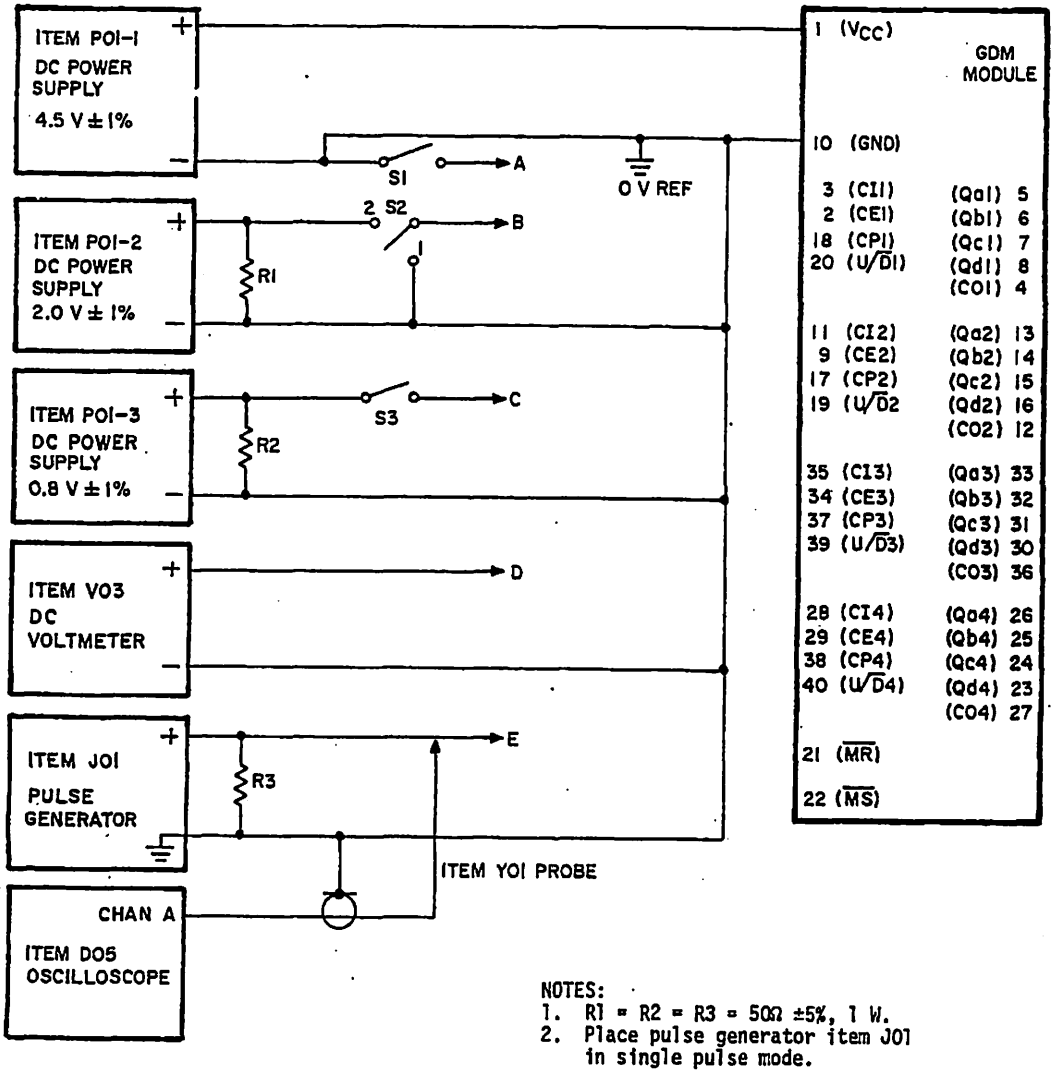
Test condition		Parameter	Connect						Initial limits 25° ±5°C Ambient		Unit	Test figure
			Pin 1 to pin	A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E18	QC4	Output voltage, high level MS(TH-L); MR(TH-H)	Open	Open	21	22	24	Open	2.4	4.5	V	8
E19	QD4	Output voltage, high level MS(TH-L); MR(TH-H)			21	22	23		2.4	4.5		
E20	CO4	Output voltage, high level MS(TH-L); MR(TH-H)			28,40 21	22	27		2.4	4.5		
F1	QA1	Output voltage, low level MR(TH-L); MS(TH-H)			22	21	5		0.00	0.40		
F2	QB1				22	21	6					
F3	QC1				22	21	7					
F4	QD1				22	21	8					
F5	CO1				22	3,20, 21	4					
F6	QA2				22	21	13					
F7	QB2				22	21	14					
F8	QC2				22	21	15					
F9	QD2				22	21	16					
F10	CO2				22	11,19 21	12					
F11	QA3				22	21	33					
F12	QB3	Output voltage, low level MR(TH-L); MS(TH-H)	Open	Open	22	21	32	Open	0.00	0.40	V	8

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TABLE V. Output voltage (low-level and high-level), functional count up and count down check (input threshold conditions) test -Continued.

Test condition	Parameter	Connect						Initial limits 25° ±5°C Ambient		Unit	Test figure
		Pin 1 to pin	A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
F13	QC3 Output voltage, low level $\overline{MR}$(TH-L); $\overline{MS}$(TH-H)	Open	Open	22	21	31	Open	0.00	0.40	V	8
F14	QD3			22	21	30					
F15	CO3			22	39,21 35	36					
F16	QA4			22	21	26					
F17	QB4			22	21	25					
F18	QC4			22	21	24					
F19	QD4			22	21	23					
F20	CO4 Output voltage, low level $\overline{MR}$(TH-L); $\overline{MS}$(TH-H)	Open	Open	22	40,21 28	27	Open	0.00	0.40	V	
G1	Functional count up	22	21	2,3 20	Open	Open	18	Note 6	Note 6		
G2				9,11, 19			17				
G3				34,35 39			37				
G4	Functional count up	22	21	28,29 40	Open		38				
H1	Functional count down	21	22	2,3	20		18				
H2		21	22	9,11	19		17				
H3		21	22	34,35	39		37				
H4	Functional count down	21	22	28,29	40	Open	38	Note 6	Note 6		8

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NOTES:
1. R1 = R2 = R3 = 50Ω ±5%, 1 W.
2. Place pulse generator item J01 in single pulse mode.

FIGURE 8. Output voltage (low-level and high-level) functional count up/down check (input threshold conditions) test circuit for table V.

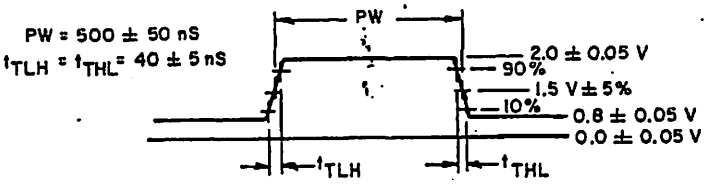
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3. Load all the outputs of the circuit under test with an output logic load as specified on figure 9. (Five separate loads are required.)
4. Switching sequence for test conditions E1-E20 and F1- F20.

   a. Initially set S2 in position 1 and S3 open. (S1 is closed.)
   b. Place S2 in position 2 and then close S3.

5. Switching sequence for test conditions G1-G4 and H1-H4.

   a. Initially close S1 and place S2 in position 2. (S3 is closed.)
   b. Open S1 and follow note 6.
   c. Input pulse at point E with E open and pulse generator J01 in one shot mode.



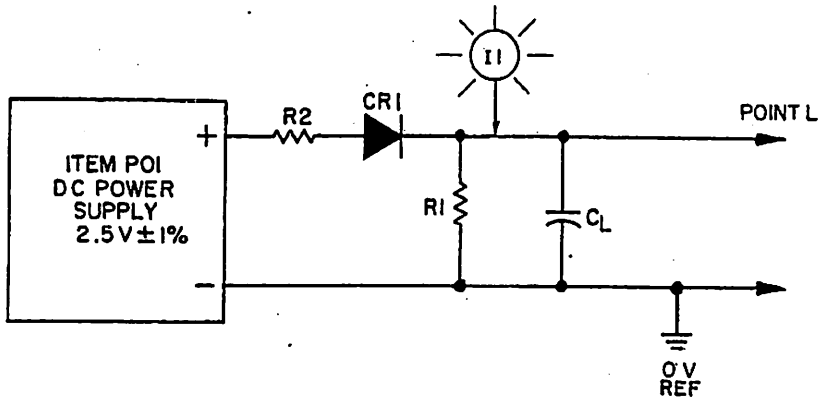
6. After following the setup procedure in note 5, the output states shall be as specified in test number 1 below. Apply one pulse to the circuit under test by triggering pulse generator item J01, and check that the output states correspond with test conditions G1-4. Repeat this for a total of 16 pulses.
7. Repeat notes 5 and 6 for test conditions H1-4.

Test conditions G1-4						
Test No.	Total pulses	CO	QD	QC	QB	QA
1	0	L	L	L	L	L
2	1	L	L	L	L	H
3	2	L	L	L	H	L
4	3	L	L	L	H	H
5	4	L	L	H	L	L
6	5	L	L	H	L	H
7	6	L	L	H	H	L
8	7	L	L	H	H	H
9	8	L	H	L	L	L
10	9	L	H	L	L	H
11	10	L	H	L	H	L
12	11	L	H	L	H	H
13	12	L	H	H	L	L
14	13	L	H	H	L	H
15	14	L	H	H	H	L
16	15	H	H	H	H	H
17	16	L	L	L	L	L

Test conditions H1-4						
Test No.	Total pulses	CO	QD	QC	QB	QA
1	0	L	H	H	H	H
2	1	L	H	H	H	L
3	2	L	H	H	L	H
4	3	L	H	H	L	L
5	4	L	H	L	H	H
6	5	L	H	L	H	L
7	6	L	H	L	L	H
8	7	L	H	L	L	L
9	8	L	L	H	H	H
10	9	L	L	H	H	L
11	10	L	L	H	L	H
12	11	L	L	H	L	L
13	12	L	L	L	H	H
14	13	L	L	L	H	L
15	14	L	L	L	L	H
16	15	H	L	L	L	L
17	16	L	H	H	H	H

FIGURE 8. Output voltage (low-level and high-level) functional count up/down check (input threshold conditions) test circuit for table V - Continued.

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NOTES:
1. CR1 = 1N4150, or equivalent.
2. C_L = 100 ±5 pF (including probe and parasitic capacitance).
3. R1 = 3.32 kΩ ±1%, 1/8 W.
4. R2 = Select value for a current flow of 10.4 mA ±1% from point L to ground.
5. I1 = A logic state indicator compatible with the logic levels and drive capabilities of the GDM module. (Required for test conditions G & H)
6. For test conditions BB and CC, C_L = 50 ±5 pF (including probe and parasitic capacitance).

FIGURE 9. GDM logic load circuit.

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TABLE VI. Power supply current test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
J1	Power supply current, high level	22	50	336	mA	10
J2	Power supply current, low level	21	50	336	mA	10

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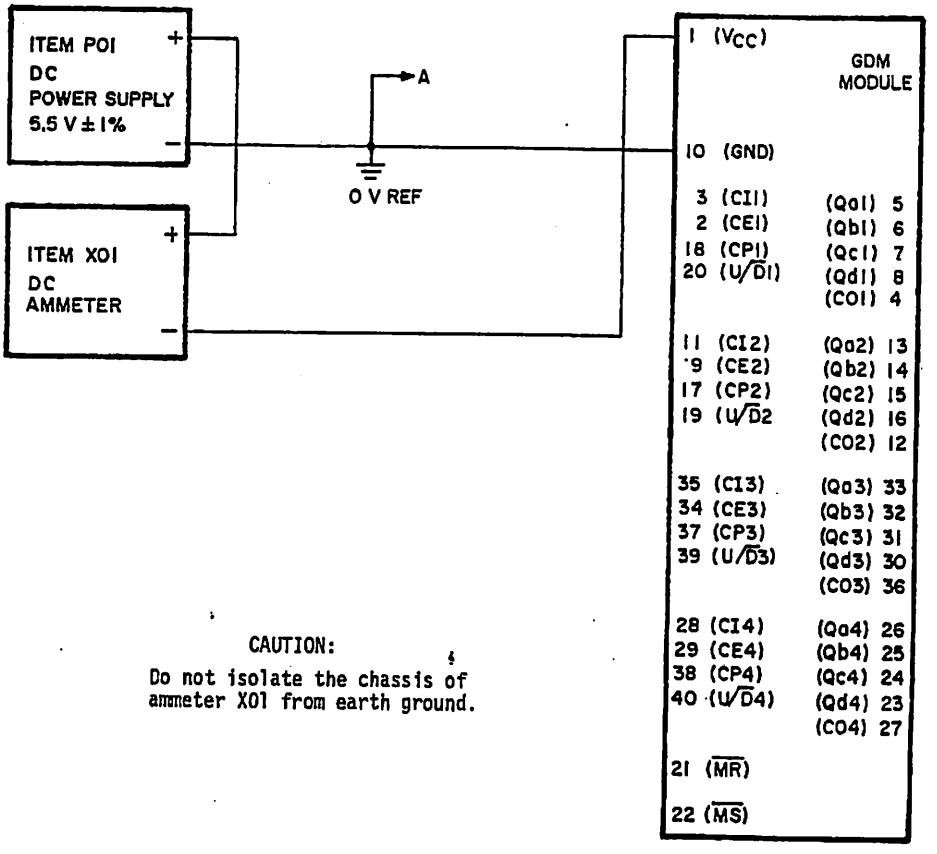


FIGURE 10. Power supply current test circuit for table VI.

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TABLE VII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) to clock test.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient				Units	Test figure
		Pin 1 to pin	A to pin	B to pin	C to pin	Pin 10 to pin	t_{THL} Min	t_{THL} Max	t_{PHL} Min	t_{PHL} Max		
K1,L1	QA1 Transition time (t_{THL}) and propagation delay time (t_{PHL})	20,21	22	18	5	Open	2	20	5	45	ns	11
K2,L2	QB1				6							
K3,L3	QC1				7							
K4,L4	QD1	20,21		18	8							
K5,L5	QA2	19,21		17	13							
K6,L6	QB2				14							
K7,L7	QC2				15							
K8,L8	QD2	19,21		17	16							
K9,L9	QA3	21,39		37	33							
K10, L10	QB3				32							
K11, L11	QC3				31							
K12, L12	QD3	21,39		37	30							
K13, L13	QA4 Transition time (t_{THL}) and propagation delay time (t_{PHL})	21,40	22	38	26	Open	2	20	5	45	ns	11

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TABLE VII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) to clock test-Continued.

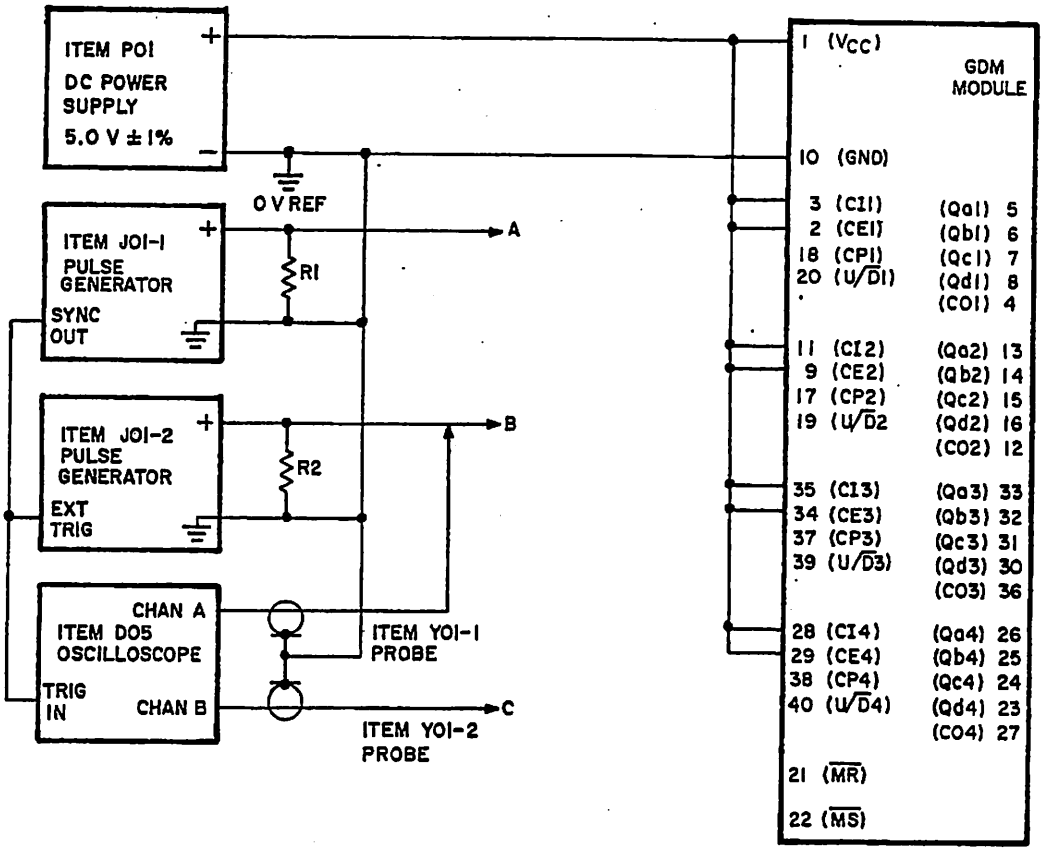
Test condition		Parameter	Connect					Initial limits 25° ±5°C ambient				Unit	Test figure
			Pin 1 to pin	A to pin	B to pin	C to pin	Pin 10 to pin	t_{THL}, t_{TLH} Min	t_{THL}, t_{TLH} Max	t_{PHL}, t_{PLH} Min	t_{PHL}, t_{PLH} Max		
K14, L14	QB4	Transition time (t_{THL}) and propagation delay time (t_{PHL})	21,40	22	38	25	Open	2	20	5	45	ns	11
K15, L15	QC4					24							
K16, L16	QD4	Transition time (t_{THL}) and propagation delay time (t_{PHL})	21,40	22	38	23	Open	2	20	5	45		
M1,N1	QA1	Transition time (t_{TLH}) and propagation delay time (t_{PLH})	22	21	18	5	20	4	40	5	40		
M2,N2	QB1					6							
M3,N3	QC1					7							
M4,N4	QD1				18	8	20						
M5,N5	QA2				17	13	19						
M6,N6	QB2					14							
M7,N7	QC2					15							
M8,N8	QD2				17	16	19						
M9,N9	QA3				37	33	39						
M10, N10	QB3					32							
M11, N11	QC3					31							
M12, N12	QD3	Transition time (t_{TLH}) and propagation delay time (t_{PLH})	22	21	37	30	39	4	40	5	40	ns	11

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TABLE VII. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) to clock test -Continued.

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient				Unit	Test figure
		Pin 1 to pin	A to pin	B to pin	C to pin	Pin 10 to pin	t_{TLH} Min	t_{TLH} Max	t_{PLH} Min	t_{PLH} Max		
M13, N13	QA4 Transition time (t_{TLH}) and propagation delay time (t_{PLH})	22	21	38	26	40	4	40	5	40	ns	11
M14, N14	QB4				25							
M15, N15	QC4				24							
M16, N16	QD4 Transition time (t_{TLH}) and propagation delay time (t_{PLH})	22	21	38	23	40	4	40	5	40	ns	11

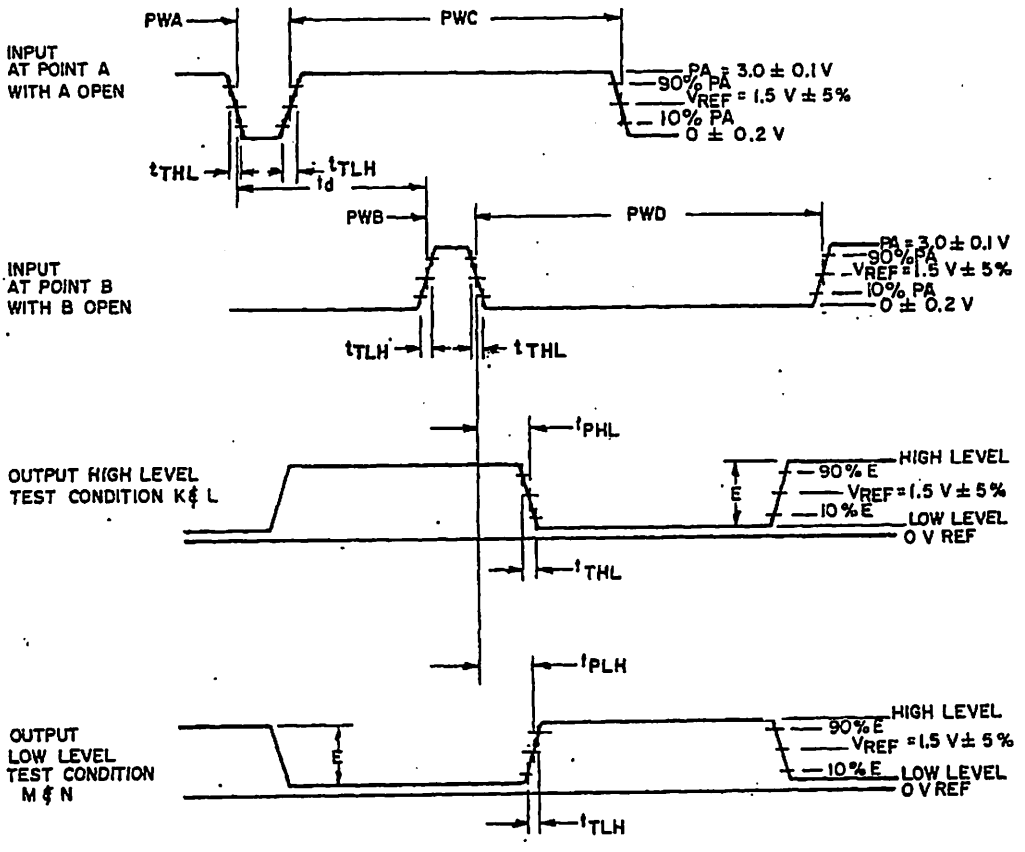
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NOTES:
1. Load all the outputs of the circuit under test with the load as specified on figure 9. (5 separate loads are required.)
2. R1 = R2 = 50Ω ±5%, 1 W.
3. Pulse A should be set up using channel B of the oscilloscope item D05 before point C is connected.

FIGURE 11. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) to clock test circuit for table VII.

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NOTE: Input pulse parameters:
PWA = 30 ±2 ns.
PWC = 170 ±20 ns.
PWB = 25 ±1 ns.
PWD = 175 ±20 ns.
t_d = 100 ±10 ns.
$t_{TLH} = t_{THL}$ = 6 ±2 ns.

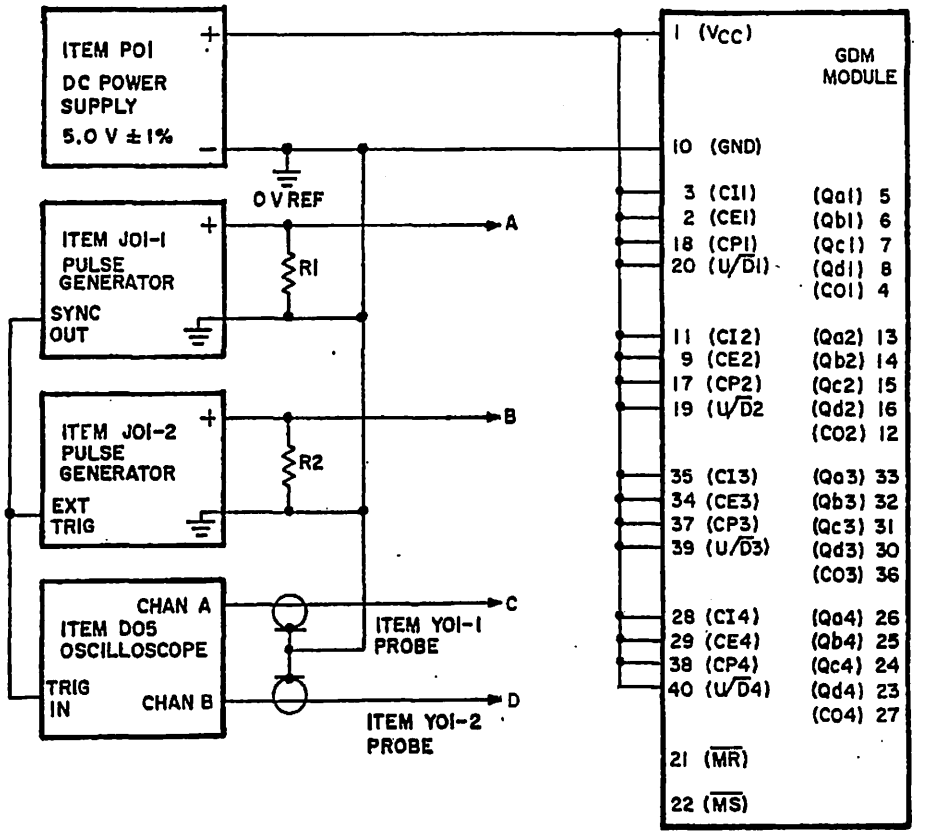
FIGURE 11. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) to clock test circuit for table VII - Continued.

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TABLE VIII. Master set and reset delay time (minimum set/reset pulse width) test.

Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient				Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Set delay time		Reset delay time			
							Min	Max	Min	Max		
P1,Q1	QA1	Set and reset delay time	21	22	22,21	5	5	35	5	40	ns	12
P2,Q2	QB1					6						
P3,Q3	QC1					7						
P4,Q4	QD1					8						
P5,Q5	QA2					13						
P6,Q6	QB2					14						
P7,Q7	QC2					15						
P8,Q8	QD2					16						
P9,Q9	QA3					33						
P10, Q10	QB3					32						
P11, Q11	QC3					31						
P12, Q12	QD3					30						
P13, Q13	QA4					26						
P14, Q14	QB4					25						
P15, Q15	QC4					24						
P16, Q16	QD4	Set and reset delay time	21	22	22,21	23	5	35	5	40	ns	12

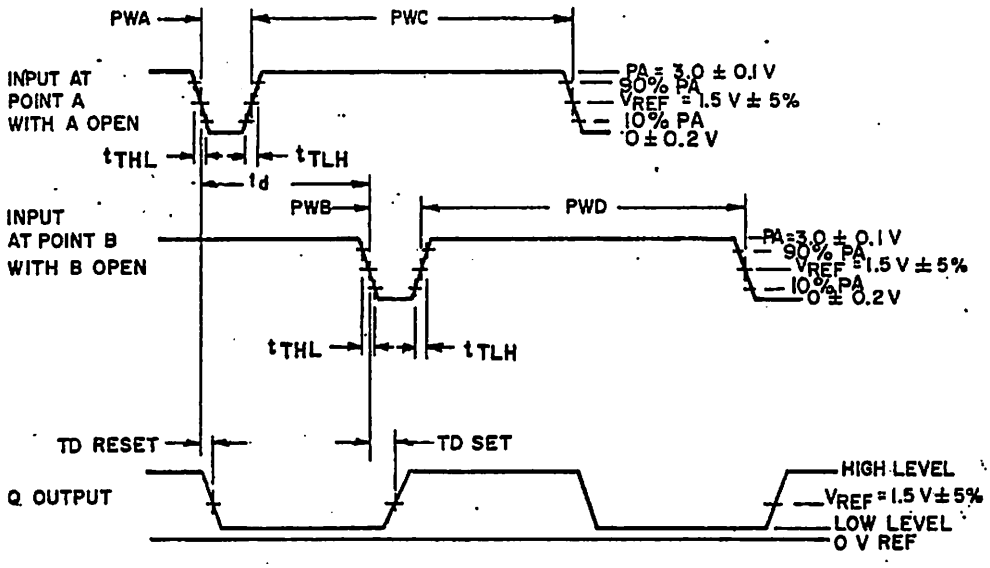
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NOTES:
1. Load all the outputs of the circuit under test with the load as specified on figure 9. (5 separate loads are required).
2. R1 = R2 = 50Ω ±5%, 1 W.
3. Pulses A to B should be setup using channels A and B of the oscilloscope before C and D are connected.

FIGURE 12. Master set and reset delay time (minimum set/reset pulse width) test circuit for table VIII.

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NOTE: Pulse parameters:
PWA = PWB = 25 ±1 ns.
PWC = PWD = 175 ±20 ns.
t_d = 100 ±10 ns.
$t_{TLH} = t_{THL}$ = 6 ±2 ns.
TD = delay time.

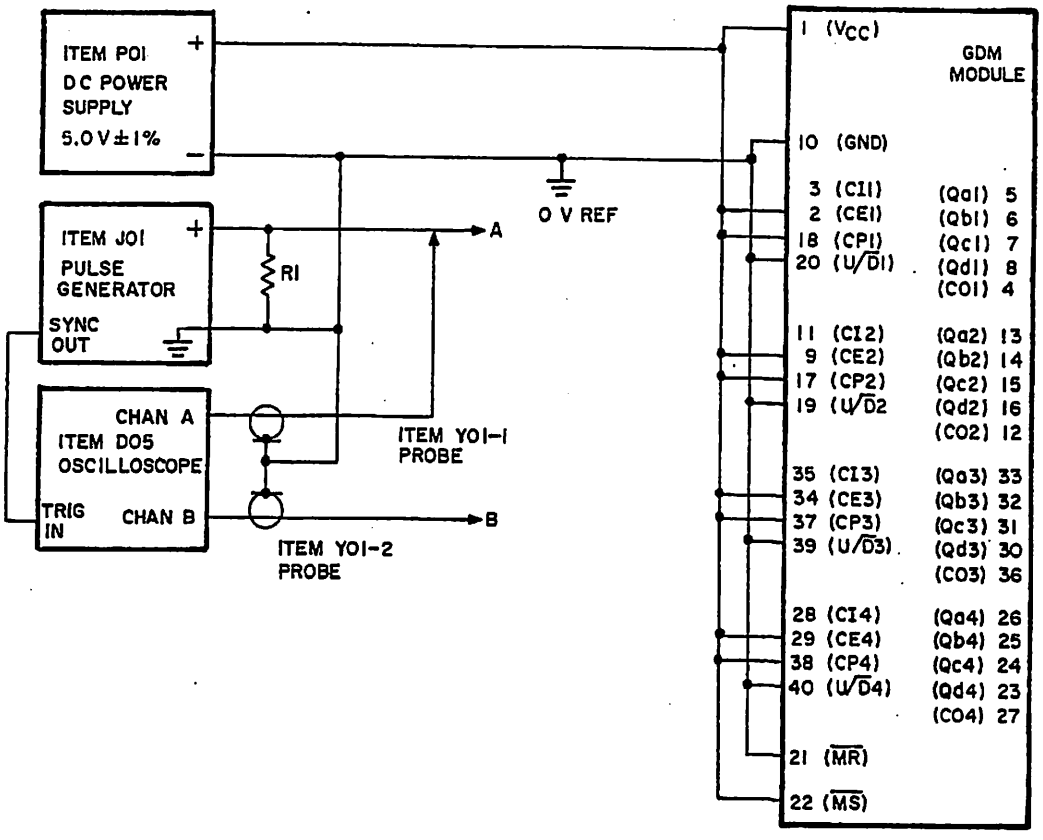
FIGURE 12. Master set and reset delay time (minimum set/reset pulse width) test circuit for table VIII - Continued.

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TABLE IX. Carry-out transition time (t_{TLH} and t_{THL}) and carry-in to carry-out propagation delay time test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient				Unit	Test figure
		A to pin	B to pin	t_{TLH}, t_{PHL}		t_{THL}, t_{PLH}			
				Min	Max	Min	Max		
R1,S1	CO1 Carry-out transition time (t_{TLH}, t_{THL})	3	4	4	40	2	20	ns	13
R2,S2	CO2	11	12						
R3,S3	CO3	35	36						
R4,S4	CO4 Carry-out transition time (t_{TLH}, t_{THL})	28	27	4	40	2	20		
T1,U1	CO1 Carry-out delay time (t_{PHL}, t_{PLH})	3	4	5	30	5	30		
T2,U2	CO2	11	12						
T3,U3	CO3	35	36						
T4,U4	CO4 Carry-out delay time (t_{PHL}, t_{PLH})	28	27	5	30	5	30	ns	13

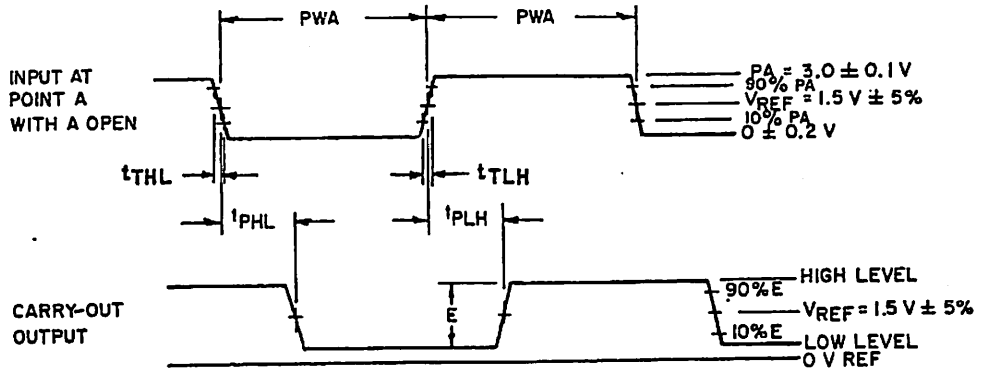
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NOTES:
1. Load Qd and CO outputs of the circuit under test with the load as specified on figure 9. (2 separate loads are required.)
2. R1 = 50Ω ±5%, 1 W.

FIGURE 13. Carry-out transition time (t_{TLH} and t_{THL}) and carry-in to carry out propagation delay time (t_{PHL} and t_{PLH}) test circuit for table IX.

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NOTE: Input pulse parameters:
PWA = 50 ±10 ns.
$t_{TLH} = t_{THL} = 6 \pm 2$ ns.

FIGURE 13. Carry-out transition time (t_{TLH} and t_{THL}) and carry-in to carry-out propagation delay time (t_{PHL} and t_{PLH}) test circuit for table IX - Continued.

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TABLE X. Carry-in, count enable, count up, and count down setup and hold time check (maximum counting frequency) test.

Test condition	Parameter	Connect: Pin 1 to pin	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
V1	Carry-in setup and hold time check	2,20	3	18	4	765	845	ns	14
V2	↓	9,19	11	17	12	↓	↓	↓	↓
V3	↓	34,39	35	37	36	↓	↓	↓	↓
V4	Carry-in setup and hold time check.	29,40	28	38	27	↓	↓	↓	↓
W1	Count enable setup and hold time check	3,20	2	18	4	↓	↓	↓	↓
W2	↓	11,19	9	17	12	↓	↓	↓	↓
W3	↓	35,39	34	37	36	↓	↓	↓	↓
W4	Count enable setup and hold time check	28,40	29	38	27	765	845	↓	↓
X1	QA1 Count up setup and hold time check	2,3	20	18	5	55	145	↓	↓
X2	QA2 ↓	9,11	19	17	13	↓	↓	↓	↓
X3	QA3 ↓	34,35	39	37	33	↓	↓	↓	↓
X4	QA4 ↓	28,29	40	38	26	55	145	↓	↓
X5	QB1 ↓	2,3	20	18	6	155	245	↓	↓
X6	QB2 ↓	9,11	19	17	14	↓	↓	↓	↓
X7	QB3 ↓	34,35	39	37	32	↓	↓	↓	↓
X8	QB4 ↓	28,29	40	38	25	155	245	↓	↓
X9	QC1 Count up setup and hold time check	2,3	20	18	7	355	445	ns	14

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TABLE X. Carry-in, count enable, count up, and count down setup and hold time check (maximum counting frequency) test -Continued.

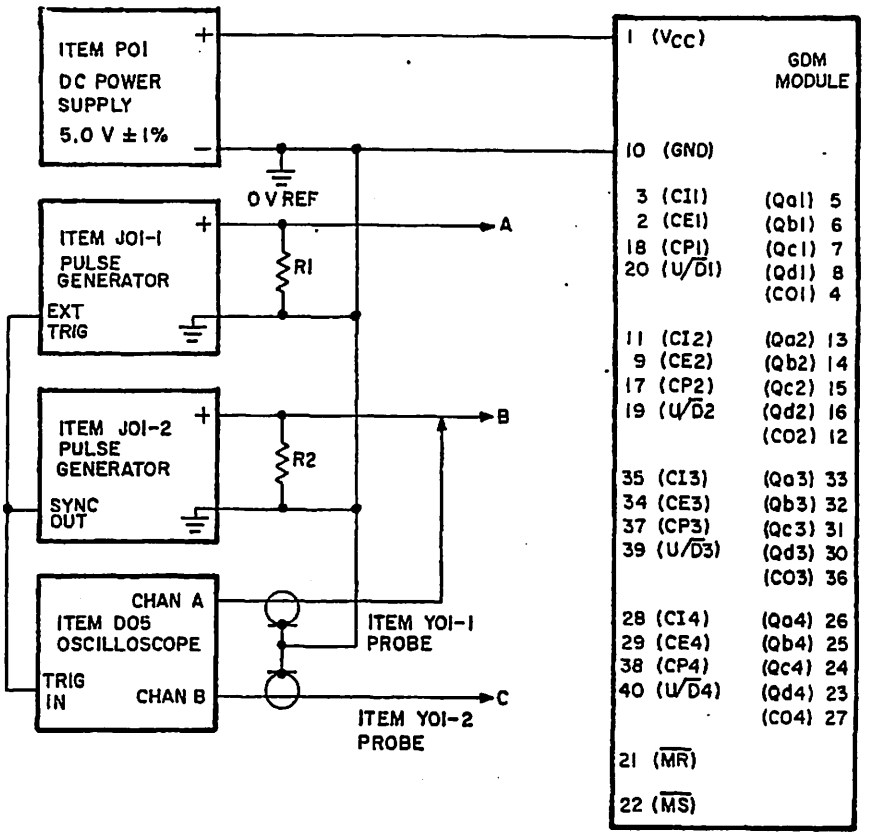
Test condition	Parameter		Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
			Pin 1 to pin	A to pin	B to pin	C to pin	Min	Max		
X10	QC2	Count up setup and hold time check	9,11	19	17	15	355	445	ns	14
X11	QC3		34,35	39	37	31	355	445		
X12	QC4		28,29	40	38	24	355	445		
X13	QD1		2,3	20	18	8	755	845		
X14	QD2		9,11	19	17	16				
X15	QD3		34,35	39	37	30				
X16	QD4	Count up setup and hold time check	28,29	40	38	23	755	845		
Y1	QA1	Count down setup and hold time check	2,3	20	18	5	55	145		
Y2	QA2		9,11	19	17	13				
Y3	QA3		34,35	39	37	33				
Y4	QA4		28,29	40	38	26	55	145		
Y5	QB1		2,3	20	18	6	155	245		
Y6	QB2		9,11	19	17	14				
Y7	QB3		34,35	39	37	32				
Y8	QB4		28,29	40	38	25	155	245		
Y9	QC1		2,3	20	18	7	355	445		
Y10	QC2		9,11	19	17	15				
Y11	QC3		34,35	39	37	31				
Y12	QC4		28,29	40	38	24	355	445		
Y13	QD1	Count down setup and hold time check	2,3	20	18	8	755	845	ns	14

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TABLE X. Carry-in, count enable, count up, and count down setup and hold time check (maximum counting frequency) test -Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		Pin 1 to pin	A to pin	B to pin	C to pin	Min	Max		
Y14	QD2 Count down setup and hold time check	9,11	19	17	16	755	845	ns	14
Y15	QD3 Count down setup and hold time check	34,35	39	37	30	755	845	ns	14
Y16	QD4 Count down setup and hold time check	28,29	40	38	23	755	845	ns	14

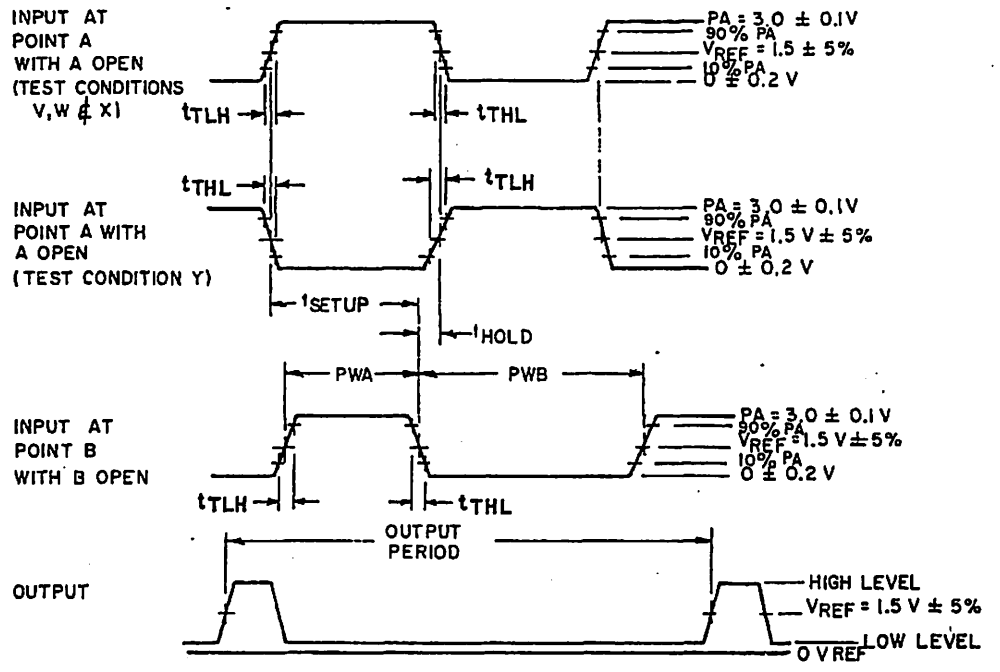
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NOTES:
1. Load all outputs of the circuit under test with the load as specified on figure 9. (5 separate loads are required.)
2. R1 = R2 = 50Ω ±5%, 1 W.
3. Pulse A should be setup using channel B of the oscilloscope item D05 before point C is connected.

FIGURE 14. Carry-in, count enable, count up, and count down setup and hold time check (maximum count frequency) test circuit for table X.

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NOTES:
1. The time scale for the inputs is not the same as that for the output.
2. The point A waveform for test condition Y can be obtained directly from the negative output of item J01-1. The baseline will have to be adjusted for 0 ±0.2 V.
3. PWA = 20 ±1 ns.
4. PWB = 30 ±1 ns.
5. t_{SETUP} = 25 ±1 ns.
   $t_{TLH} = t_{THL}$ = 6 ±2 ns.
   t_{HOLD} = 2 +0, -2 ns.

FIGURE 14. Carry-in, count enable, count up, and count down setup and hold time check (maximum count frequency) test circuit for table X - Continued.

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TABLE XI. Isolation test.

Test condition	Parameter	Connect megohm bridge item RO6		Initial limits 25° ±5°C ambient		Unit	Test figure
		+ Terminal	-Terminal	Min	Max		
Z	Isolation	All pins	*Module fin	10	∞	MΩ	

WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "Discharge" before connecting or disconnecting the test leads.

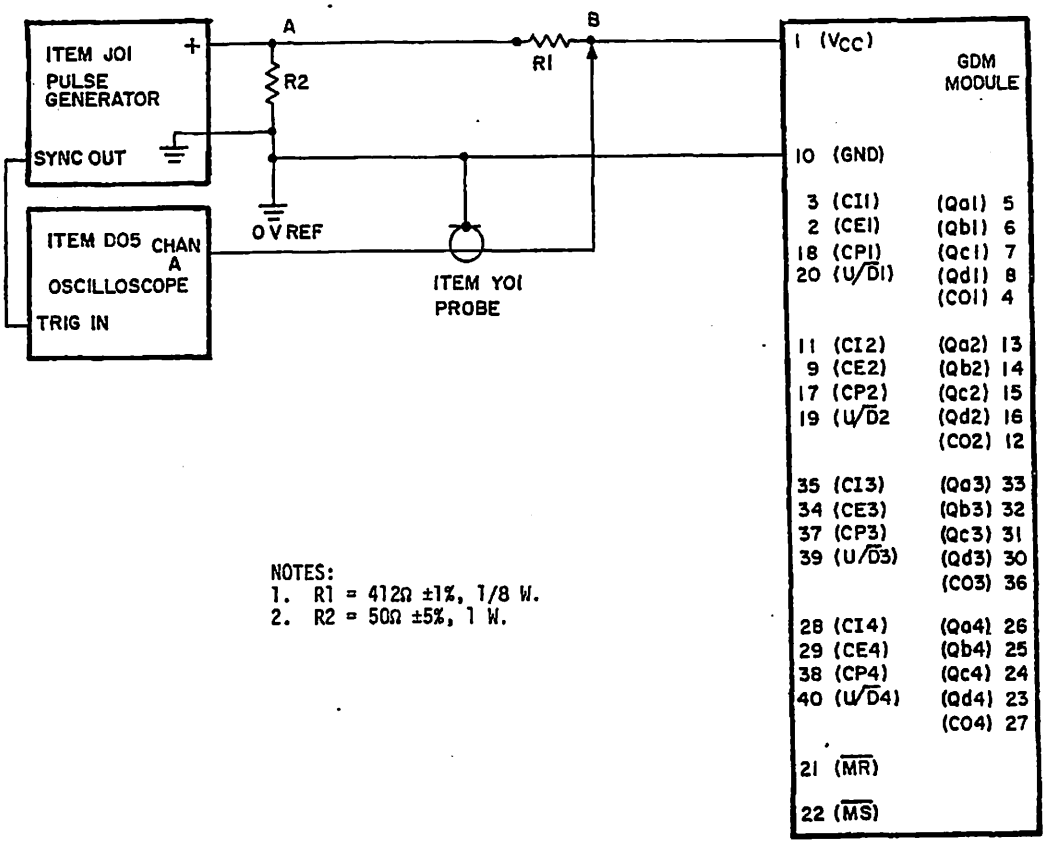
*Penetrate the anodized finish on the module fin with a sharply pointed probe. Connect the negative lead of the megohm bridge to this probe. The megohm bridge output shall be set at 100 ± 10 Vdc.

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TABLE XII. Power supply transient rise time test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
AA	Power supply transient rise time	2.5	7.5	µs	15

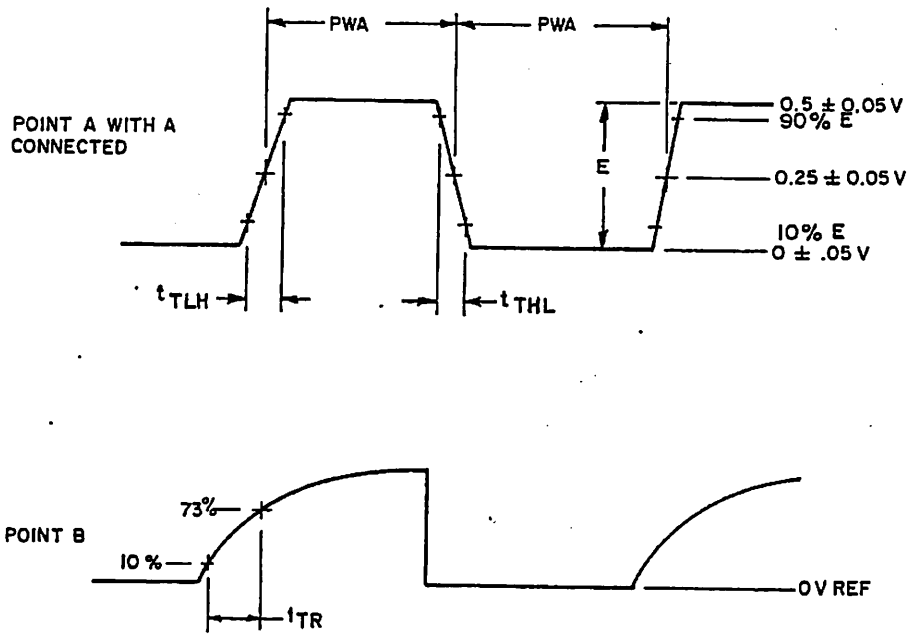
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NOTES:
1. R1 = 412Ω ±1%, 1/8 W.
2. R2 = 50Ω ±5%, 1 W.

FIGURE 15. Power supply transient rise time test circuit for table XII.

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NOTES:
1. PWA = 30 ±µs.
2. $t_{TLH} = t_{THL} = 50 \pm 5$ ns.

FIGURE 15. Power supply transient rise time test circuit for table XII - Continued.

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TABLE XIII. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Connect: D to pin	Connect: E to pin	Connect: Pin to pin	Initial limits 25° ±5°C ambient: Min	Initial limits 25° ±5°C ambient: Max	Unit	Test figure
BB1	QA1 Crosstalk, high level	3,9,11,20, 21,28,29, 34,35	22	17,37, 38	5	2,18	13 to 19 33 to 39 26 to 40	0.0	0.4	V	16
BB2	QB1				6						
BB3	QC1				7						
BB4	QD1				8						
BB5	CO1	3,9,11,20, 21,28,29, 34,35		17,37, 38	4	2,18	13 to 19 33 to 39 26 to 40				
BB6	QA2	2,3,11,19, 21,28,29,34, 35		18,37, 38	13	9,17	5 to 20 33 to 39 26 to 40				
BB7	QB2				14						
BB8	QC2				15						
BB9	QD2				16						
BB10	CO2	2,3,11,19, 21,28,29, 34,35		18,37, 38	12	9,17	5 to 20 33 to 39 26 to 40				
BB11	QA3	2,3,9,11, 21,28,29, 35,39		17,18, 38	33	34,37	5 to 20 13 to 19 26 to 40				
BB12	QB3 Crosstalk, high level	2,3,9,11, 21,28,29, 35,39	22	17,18, 38	32	34,37	5 to 20 13 to 19 26 to 40	0.0	0.4	V	16

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TABLE XIII. Output crosstalk (high-level and low-level) test -Continued.

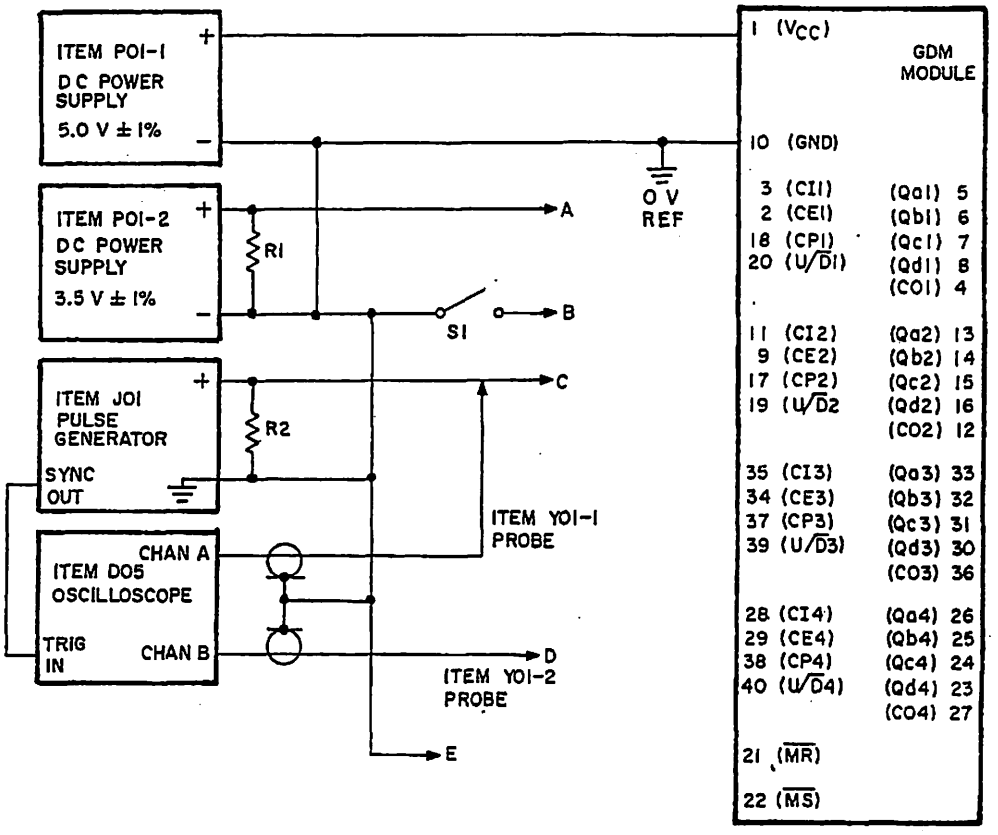
Test condition	Parameter	Connect: A to pin	B to pin	C to pin	D to pin	E to pin	Pin to pin	Initial limits 25° ±5°C ambient: Min	Max	Unit	Test figure
BB13	QC3 Crosstalk, high level	2,3,9,11, 21,28,29, 35,39	22	17,18, 38	31	34,37	5 to 20 13 to 19 26 to 40	0.0	0.4	V	16
BB14	QD3	2,3,9,11, 21,28,29, 35,39		17,18, 38	30		5 to 20 13 to 19 26 to 40				
BB15	CO3	2,3,9,11, 21,28,29, 35,39		17,18, 38	36	34,37	5 to 20 13 to 19 26 to 40				
BB16	QA4	2,3,9,11, 21,28,34, 35,40		17,18, 37	26	29,38	5 to 20 13 to 19 33 to 39				
BB17	QB4				25						
BB18	QC4				24						
BB19	QD4				23						
BB20	CO4 Crosstalk, high level	2,3,9,11, 21,28,34, 35,40	22	17,18, 37	27	29,38	5 to 20 13 to 19 33 to 39				
CC1	QA1 Crosstalk, low level	9,11,22,28, 29,34,35	21	17,37, 38	5	2,3, 18,20	13 to 19 33 to 39 26 to 40				
CC2	QB1				6						
CC3	QC1				7						
CC4	QD1				8						
CC5	CO1 Crosstalk, low level	9,11,22,28, 29,34,35	21	17,37, 38	4	2,3, 18,20	13 to 19 33 to 39 26 to 40	0.0	0.4	V	16

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TABLE XIII. Output crosstalk (high-level and low-level) test -Continued.

Test condition	Parameter	Connect						Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Pin to pin	Min	Max		
CC6	QA2 Crosstalk, low level	2,3,22,28, 29,34,35	21	18,37, 38	13	9,11, 17,19	5 to 20 33 to 39 26 to 40	0.0	0.4	V	16
CC7	QB2				14						
CC8	QC2				15						
CC9	QD2				16						
CC10	CO2	2,3,22,28, 29,34,35		18,37, 38	12	9,11, 17,19	5 to 20 33 to 39 26 to 40				
CC11	QA3	2,3,9,11, 22,28,29		17,18, 38	33	34,35 37,39	5 to 20 13 to 19 26 to 40				
CC12	QB3				32						
CC13	QC3				31						
CC14	QD3				30						
CC15	CO3	2,3,9,11, 22,28,29		17,18, 38	36	34,35, 37,39	5 to 20 13 to 19 26 to 40				
CC16	QA4	2,3,9,11, 22,34,35		17,18, 37	26	28,29, 38,40	5 to 20 13 to 19 33 to 39				
CC17	QB4				25						
CC18	QC4				24						
CC19	QD4				23						
CC20	CO4 Crosstalk, low level	2,3,9,11, 22,34,35	21	17,18, 37	27	28,29, 38,40	5 to 20 13 to 19 33 to 39	0.0	0.4	V	16

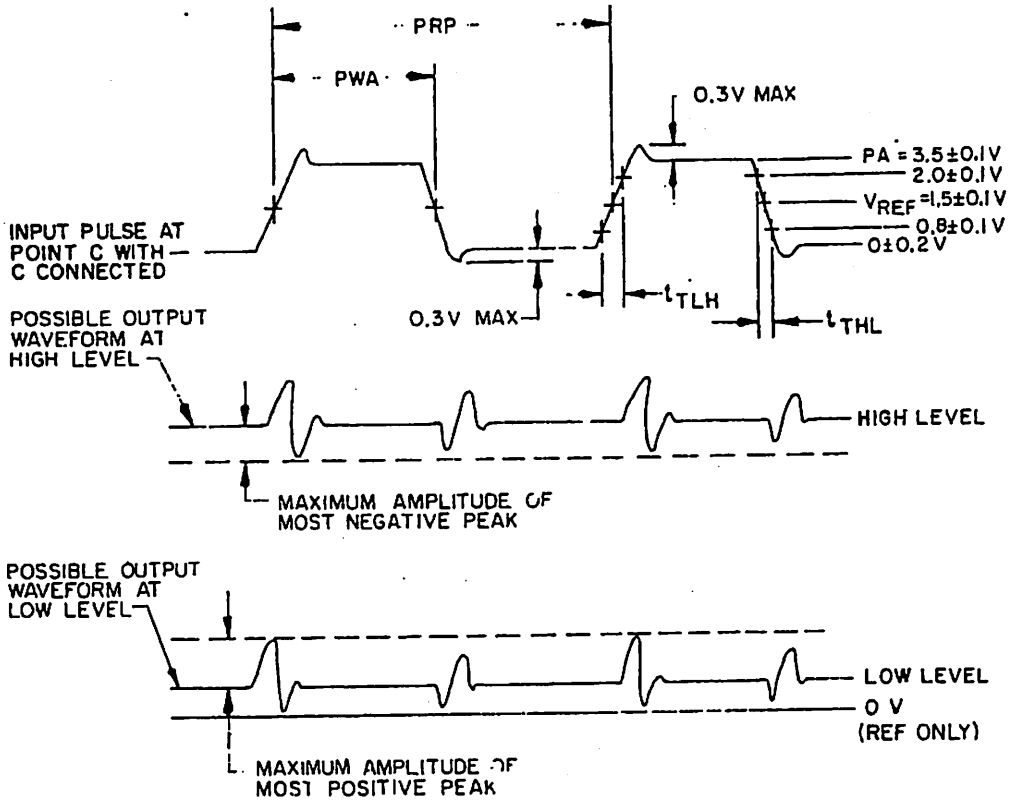
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. After making all circuit connections for a test condition close S1 for at least 100 ns, then open S1 and connect the pins at B to A.
3. Load all outputs with the load specified in figure 9, except C_L = 50 ±5 pF.
4. Verify that all other counter outputs, except the one under test, changes states simultaneously and to the same state.

FIGURE 16. Output crosstalk (high-level and low-level) test circuit for table XIII.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 100 ±10 ns.
   PRP = 200 ±20 ns.
2. The lead length of electrical conductors between terminator resistor R1 and input pins; probes YO1-1, YO1-2 to input and output pins; load circuits to output pins; and ground leads of probes YO1-1, YO1-2 to test fixture and module ground pins, shall be less than 2 inches (50.8 mm).

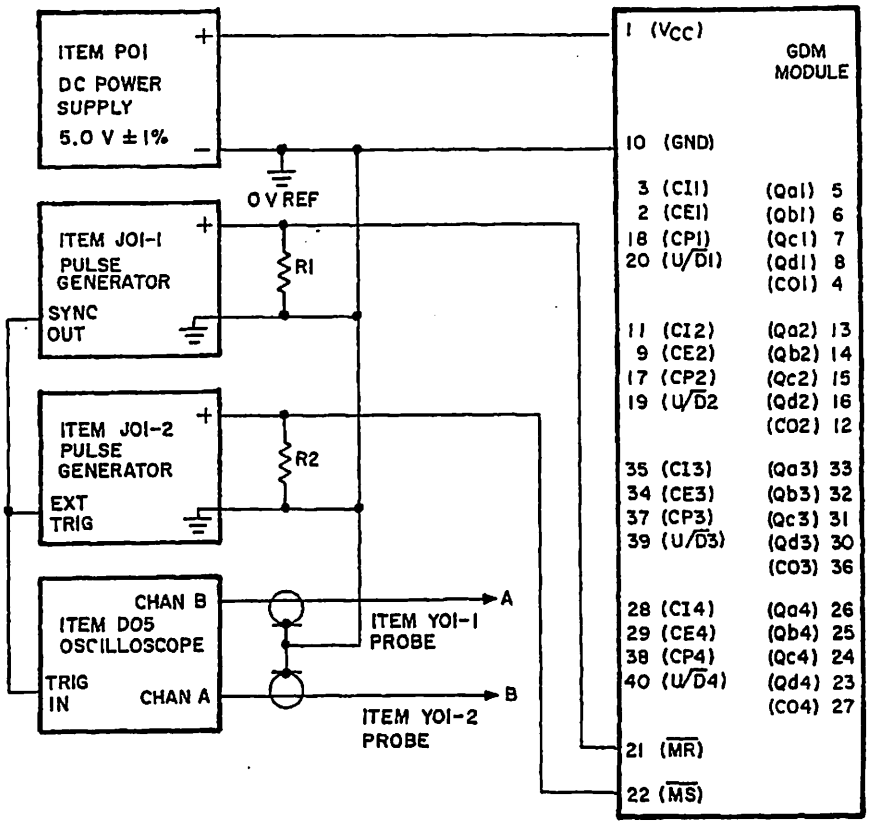
FIGURE 16. Output crosstalk (high-level and low-level) test circuit for table XIII - Continued.

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TABLE XIV. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
DD	Power supply transient amplitude	1	0	400	mV	17

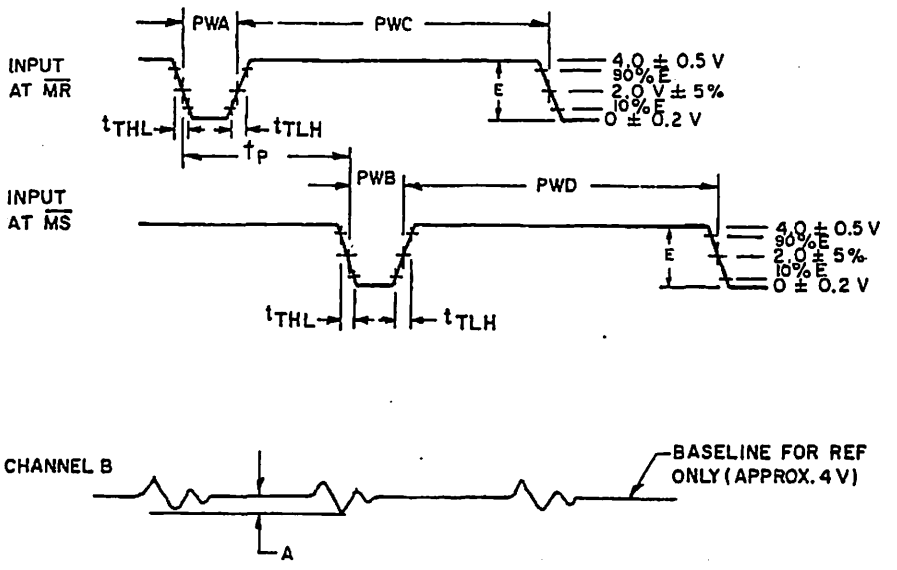
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:32Z
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NOTES:
1. R1 = R2 = 50Ω ±5%, 1 W.
2. Inputs at $\overline{MR}$ and $\overline{MS}$ should be set up using channels A and B of the oscilloscope before A and B are connected.
3. Both input waveforms should be viewed to determine where the most negative excursion occurs.

FIGURE 17. Power supply transient amplitude test circuit for table XIV.

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NOTE: "A" is measured from baseline to the most negative excursion of the waveform.
PWA = PWB = 25 ±2 ns.
PWC = PWD = 175 ±20 ns.
t_p = 100 ±10 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.

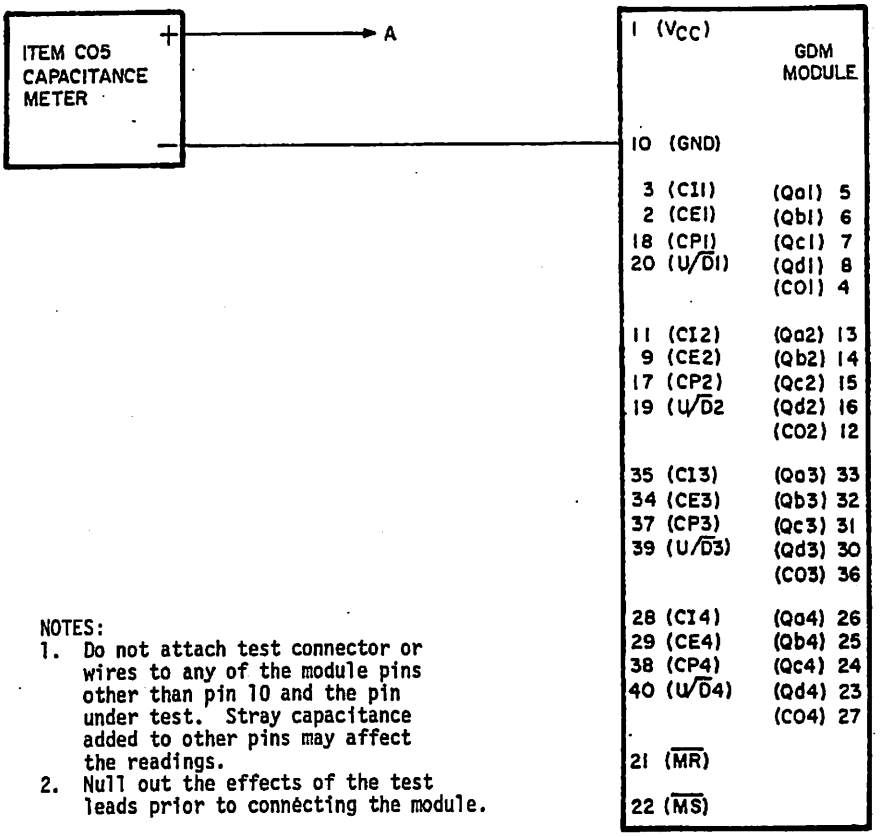
FIGURE 17. Power supply transient amplitude test circuit for table XIV - Continued.

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TABLE XV. Input capacitance test.

Test condition	Parameter	Connect	Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	Min	Max		
EE1	$\overline{MR}$ Input capacitance	21	16	96	pF	18
EE2	$\overline{MS}$	22	16	96		
EE3	CP1	18	2	15		
EE4	CP2	17				
EE5	CP3	37				
EE6	CP4	38				
EE7	CI1	3				
EE8	CE1	2				
EE9	U/$\overline{D1}$	20				
EE10	CI2	11				
EE11	CE2	9				
EE12	U/$\overline{D2}$	19				
EE13	CI3	35				
EE14	CE3	34				
EE15	U/$\overline{D3}$	39				
EE16	CI4	28				
EE17	CE4	29				
EE18	U/$\overline{D4}$ Input capacitance	40	2	15	pF	18

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

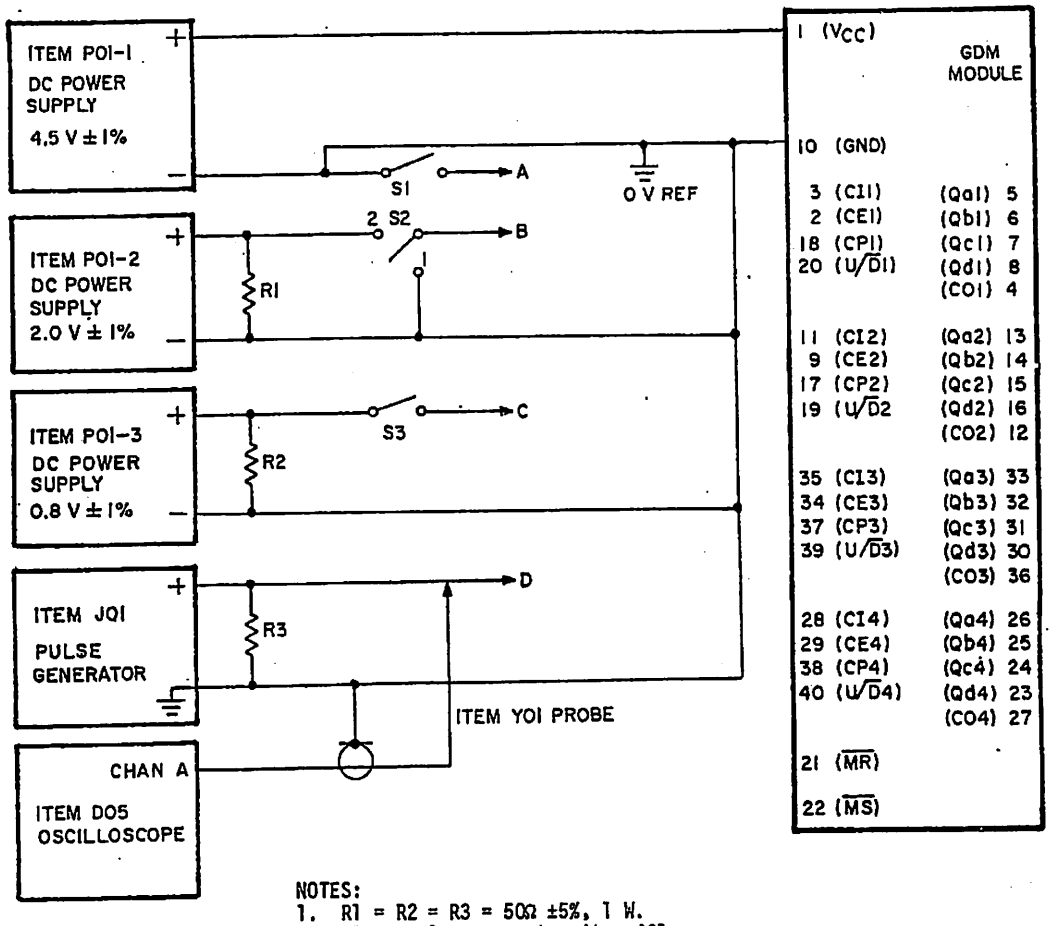
FIGURE 18. Input capacitance test circuit for table XV.

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TABLE XVI. Functional count up and count down check (input threshold conditions).

Test condition	Parameter	Connect					Initial limits 25° ±5°C ambient		Unit	Test figure
		Pin 1 to pin	A to pin	B to pin	C to pin	D to pin	Min	Max		
FF1	Functional count up	22	21	2,3, 20	Open	18	Note 5	Note 5		19
FF2				9,11, 19		17				
FF3				34,35, 39		37				
FF4	Functional count up	22	21	28,29, 40	Open	38				
GG1	Functional count down	21	22	2,3	20	18				
GG2		21	22	9,11	19	17				
GG3		21	22	34,35	39	37				
GG4	Functional count down	21	22	28,29	40	38	Note 5	Note 5		19

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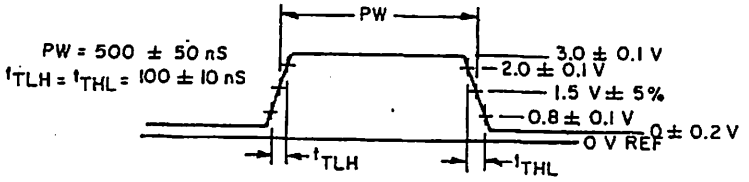
NOTES:
1. R1 = R2 = R3 = 50Ω ±5%, 1 W.
2. Place pulse generator item J01 in single pulse mode.

FIGURE 19. Functional count up and count down check (input threshold conditions) test circuit for table XVI.

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3. Load all the outputs of the circuit under test with an output logic load as specified in figure 9. (Five separate loads are required.)
4. Switching sequence for test conditions FF1-4 and GG1-4.

   a. Initially close S1 and place S2 in position 2. (S3 is closed.)
   b. Open S1 and follow note 4.
   c. Input pulse at point D with D open and pulse generator J01 in one shot mode.



5. After following the setup procedure in note 4, the output states shall be as specified in test number 1 below. Apply one pulse to the circuit under test by triggering pulse generator item J01, and check that the output states correspond with test conditions FF1-4. Repeat this for a total of 16 pulses.
6. Repeat notes 4 and 5 for test conditions GG1-4.

Test conditions FF1-4						
Test No.	Total pulses	CO	QD	QC	QB	QA
1	0	L	L	L	L	L
2	1	L	L	L	L	H
3	2	L	L	L	H	L
4	3	L	L	L	H	H
5	4	L	L	H	L	L
6	5	L	L	H	L	H
7	6	L	L	H	H	L
8	7	L	L	H	H	H
9	8	L	H	L	L	L
10	9	L	H	L	L	H
11	10	L	H	L	H	L
12	11	L	H	L	H	H
13	12	L	H	H	L	L
14	13	L	H	H	L	H
15	14	L	H	H	H	L
16	15	H	H	H	H	H
17	16	L	L	L	L	L

Test conditions GG1-4						
Test No.	Total pulses	CO	QD	QC	QB	QA
1	0	L	H	H	H	H
2	1	L	H	H	H	L
3	2	L	H	H	L	H
4	3	L	H	H	L	L
5	4	L	H	L	H	H
6	5	L	H	L	H	L
7	6	L	H	L	L	H
8	7	L	H	L	L	L
9	8	L	L	H	H	H
10	9	L	L	H	H	L
11	10	L	L	H	L	H
12	11	L	L	H	L	L
13	12	L	L	L	H	H
14	13	L	L	L	H	L
15	14	L	L	L	L	H
16	15	H	L	L	L	L
17	16	L	H	H	H	H

FIGURE 19. Functional count up and count down check (input threshold conditions) test circuit for table XVI - Continued.

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